Predicting Loan Default in Peer-to-Peer Lending: A Comparative Machine Learning Study of Logistic Regression, Random Forest, and Neural Networks under EU AI Act Governance

Abstract

This paper includes comprehensive comparative research on loan default prediction algorithms utilizing a refined sample of 40,000 LendingClub records. First, building a clear Logistic Regression baseline which has firm coefficient interpretability and then assess elaborate architectures, namely, Random Forest and a deep Multi-Layer Perceptron Neural Network. Then, the application of a preprocessing pipeline handled class imbalance using class-weighted model training, categorical encoding, and standardization of features. The models were compared based on the ROC-AUC , F1-scores, and confusion matrices, in a purely operationalized way such that the requirements of the EU AI Act are adhered to. The results indicate that the Neural Network has better discriminative performance, whereas the Logistic Regression baseline has invaluable regulatory transparency and auditability.To address the sophisticated mathematical and legal challenges of model interpretability, this study introduces a two-leaderboard approach ranking models based on their predictive accuracy and interpretability as a framework of structural governance.

Introduction

Fintech-driven peer-to-peer (P2P) lending is transforming consumer modern credit risk management especially in the financial sector [1]. According to Stripe, the worldwide fintech lending industry is predicted to rise from $200 billion in 2022 to over $1 .8 trillion by 2032 [2]. This growth demands that the dependence of traditional credit scoring methods on limited variables like FICO scores, should be replaced by sophisticated machine learning (ML) algorithms. These algorithms can process hundreds of borrower-specific features to produce highly discriminative default forecasts, significantly outperforming classical underwriting processes [3, 4]. Consequently, the Lending Club platform has become a foundational dataset in credit risk ML research, offering comprehensive loan-level and behavioural metadata across diverse macroeconomic cycles [5, 6].

This study is driven by three primary motivations. First, a persistent empirical gap exists between models maximizing predictive accuracy via opaque architectures and those prioritizing structural interpretability. This difference poses practical challenges for lenders, regulators, and consumers. Second, the EU AI Act (2024) classifies algorithmic credit scoring as high-risk, mandating strict systemic transparency, auditability, and human oversight. This classification creates an urgent demand for governance frameworks balancing predictive execution with legal compliance [7]. Third, existing methodological literature rarely evaluates systematic models in a unified classical statistical inference, ensemble learning, and deep neural networks within a single reproducible pipeline on the Lending Club dataset [8].

Research Questions

This study investigates the predictive, interpretative, and regulatory implications of applying different machine learning algorithms to peer-to-peer(P2P) lending default prediction. Specifically, the research addresses the following questions:

i. Which ML architecture achieves the highest discriminative accuracy?
ii. What is the explanatory cost of exceeding Logistic Regression baselines?
iii. What are the most potent, theoretically consistent default predictors?
iv. How can a dual governance structure ensure both accuracy and EU AI Act compliance?

Related Work

2.1 Foundational Credit Risk Modelling

Credit risk modelling originated with Altman's Z-score, which showed that weighted financial ratios could separate defaulting from solvent firms[9].In consumer lending, Logistic Regression became the dominant industry standard because it provides calibrated probabilities, coefficient-level interpretability, and regulatory transparency. Thomas et al. later formalized operational scorecard frameworks widely used in banking practice[10].However, empirical studies by Baesens et al. demonstrated that nonlinear machine learning models such as Neural Networks and SVMs often outperform Logistic Regression in predictive accuracy, albeit with reduced interpretability[4]. This established the modern accuracy–transparency trade-off and motivates the present P2P lending default prediction framework integrating classical and machine

2.2 Machine Learning for P2P Lending Default Prediction

Early empirical work on LendingClub data by Emekter et al . applied Logistic Regression and survival models, identifying platform credit grade, borrower FICO score, debt-to-income ratio, and revolving utilization as key default drivers, while noting residual risk beyond internal grades exploitable by external models[11]. Serrano-Cinca et al. extended this by jointly analysing interest-rate assignment and default, showing through correlation and information-gain analysis that platform pricing algorithms mispriced the riskiest borrower segments[12]. Malekipirbazari and Aksakalli demonstrated the strong predictive power of Random Forest on LendingClub data, achieving ROC-AUC above 0.90 on balanced samples and confirming that traditional FICO retained independent explanatory strength alongside platform grades[13]. From a structural perspective, Milne framed P2P lending within financial disintermediation, arguing that platform lenders face heightened adverse selection due to the absence of qualitative relationship-banking information available to traditional institutions[14]. Together, these studies establish LendingClub as a benchmark dataset and motivate advanced machine-learning approaches for improved risk modelling.

2.3. Ensemble methods and Neural Networks.

Ensemble learning advanced credit risk modelling by aggregating multiple decision trees into Random Forests, reducing variance and improving robustness to noise and non-stationarity in retail financial data [13]. Parallel research showed that deep neural networks, particularly multilayer perceptrons, can model complex nonlinear interactions among borrower variables that tree methods may miss [15, 16]. While early neural applications suffered severe overfitting, modern regularization techniques such as dropout, batch normalization, and adaptive optimizers enabled performance comparable to or exceeding ensembles [16]. Despite these gains, neural models impose substantial interpretability costs, as their opaque internal structure limits transparency and historically constrained deployment in regulated credit underwriting contexts [15].

2.4 Class Imbalance in Default Prediction

The LendingClub platform historical origination data shows that the defaults rates are inherently moving between 15 and 25, depending on the year of issue and the macroeconomic environment

[17]. The resulting effect of this extreme distribution of classes is a systematic classification bias, in which the overall precision is maximized by the machine learning algorithms that trivially clustered the majority class (non-default) and often failed catastrophically to identify the minority class (actual defaults) [18].

To counter this bias mathematically, Chawla *et al*. proposed the Synthetic Minority Over-sampling Technique (SMOTE), which quickly causes overfitting instead of merely replicating the existing minority examples [18]. SMOTE works in the geometric feature space by interpolating new artificial examples of the minority classes along the line segments between the k-nearest example minority neighbors [18]. Their surveys also unanimously found the strongest predictive results with ensemble methods applied in conjunction with stringent SMOTE resampling techniques, in highly imbalanced financial tasks [18].

2.5 Regulatory Dimensions: Model Governance and EU AI Act.

The regulatory environment of algorithmic underwriting is radically changing, with the adoption of the EU AI Act being one of the major drivers towards these changes. The legislation is an extensive legal framework that applies to artificial intelligence[7]. Most importantly, the Act explicitly categorizes credit scoring systems and algorithms applied to assess the creditworthiness of natural persons as high-risk AI processes in Annex III [7]. This label requires compliance with transparency procedures, ongoing audits of algorithmic robustness, extensive technical documentation, and verifiable human-in-the-loop supervision systems[7]. The short-term operational implication on financial institutions is that “black-box” predictive systems such as highly optimized Neural Networks should be supplemented with transparent parallel baseline models, which act as regulatory proxies[7]. Hull’s approach focused on the critical derivation of t-statistics and p-values as well as odds ratios to develop legally sound credit risk models that fully satisfy regulatory requirements while applying Logistic Regression[19].

2.6 Recent Advances and Ayari et al. (2025)

Ayari *et al.* provide a recent review of machine learning approaches to fintech credit applications by recommending unified model governance frameworks[8]. Their meta-analytic study conclusively reveals that no model architecture can be significantly better than the other in all financial data or measures of evaluation, which makes the comparative approach of the

methodology used in this study highly motivating [8]. Ayari *et al.* insist on the need to create two-optimalization environment where institutions will not only base their considerations on Receiver Operating Characteristic (ROC) dynamics but also on their ability to endure strict regulatory compliance and fairness audit[8].

3. Methodology

3.1 Dataset Description

This empirical research utilizes the publicly available LendingClub dataset from Kaggle, specifically focusing on the accepted loans repository containing 2.26 million historical records spanning 2007 to 2018[17]. This expansive chronological window is crucial for financial modelling as it captures diverse macroeconomic phases, including the post-2008 financial crisis recovery and subsequent economic expansions. To ensure computational tractability and algorithm stability within Python, a rigorously stratified random sample of 40,000 records was extracted, perfectly preserving original temporal and class distributions. The target dependent variable was engineered using a strict binary definition: "Charged Off" loans were coded as default events (1), and "Fully Paid" loans as non-defaults (0). Ambiguous intermediate statuses were systematically excluded to prevent data leakage[13]. This refined sample exhibited an initial default rate of 21.8%, confirming structural class imbalance[18].

3.2 Exploratory Data Analysis (EDA)

LendingClub data (151 features) were reduced to 35 predictors using EDA and heuristic filtering. Features with over 30% missing values were removed to limit imputation bias. Post-origination variables (e.g., recoveries, late fees) were excluded to prevent information leakage and ensure realistic default prediction at underwriting.

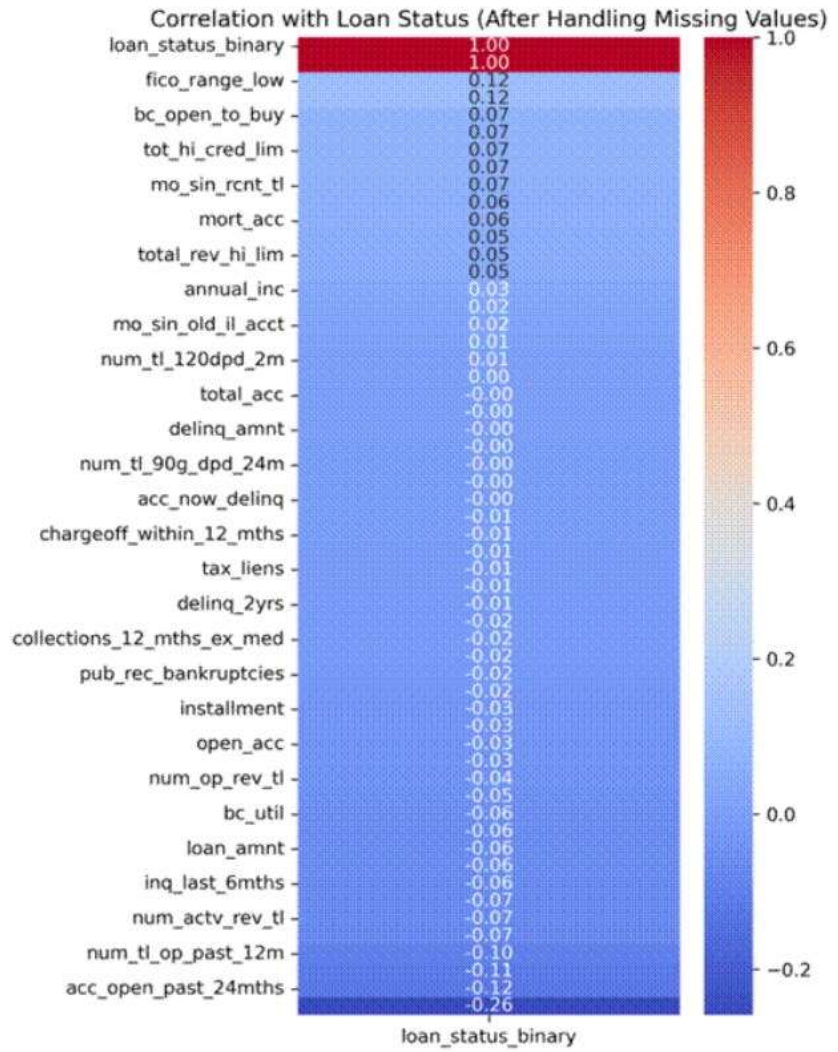

Figure 1: Correlation of numeric LendingClub features with loan default status after median imputation

Borrower credit quality and capacity variables, such as higher FICO scores and available credit versus high utilization and recent credit activity, are most predictive of loan outcomes and highlight the need for multivariate ML models to capture combined risk effects.

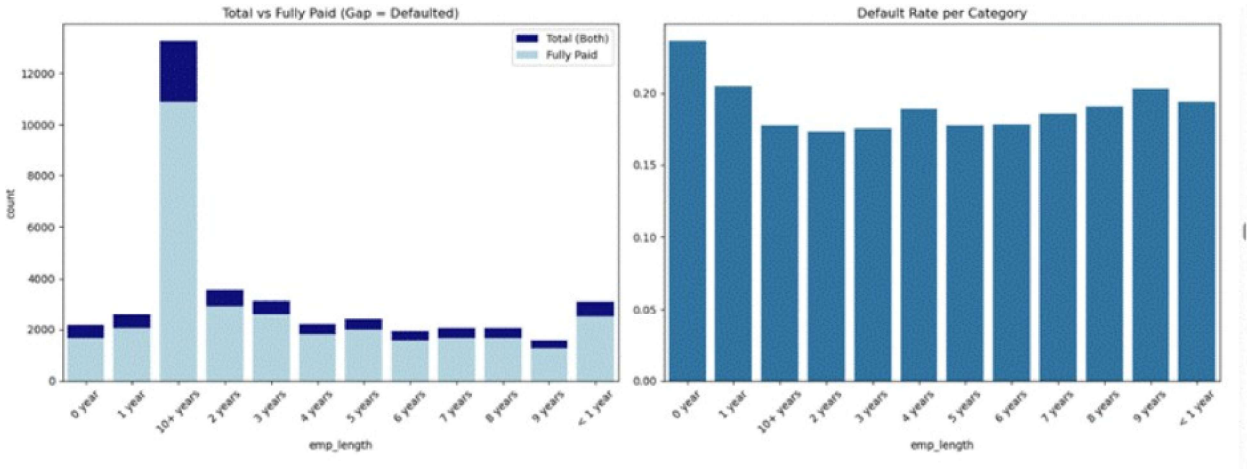

Figure 2: Loan distribution and default rate by borrower employment length (emp_length).

Default rates vary moderately across employment tenure, with shorter job histories showing Oslightly higher risk. Longer employment duration aligns with more stable repayment, supporting its role as a behavioural credit risk indicator.

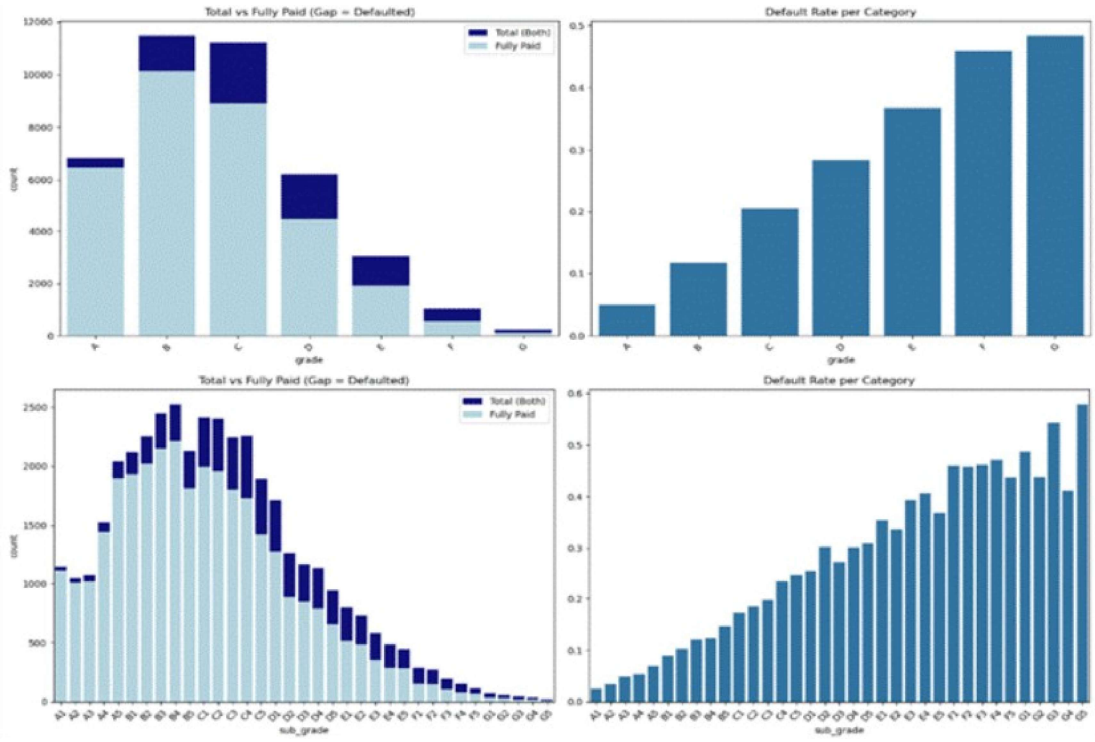

Figure 3: Loan distribution and default rate across LendingClub credit grade and sub-grade categories.

LendingClub credit grades show a strong monotonic rise in default risk from A to G (and A1-G5), confirming credit grade and sub-grade as key predictors in default modelling.

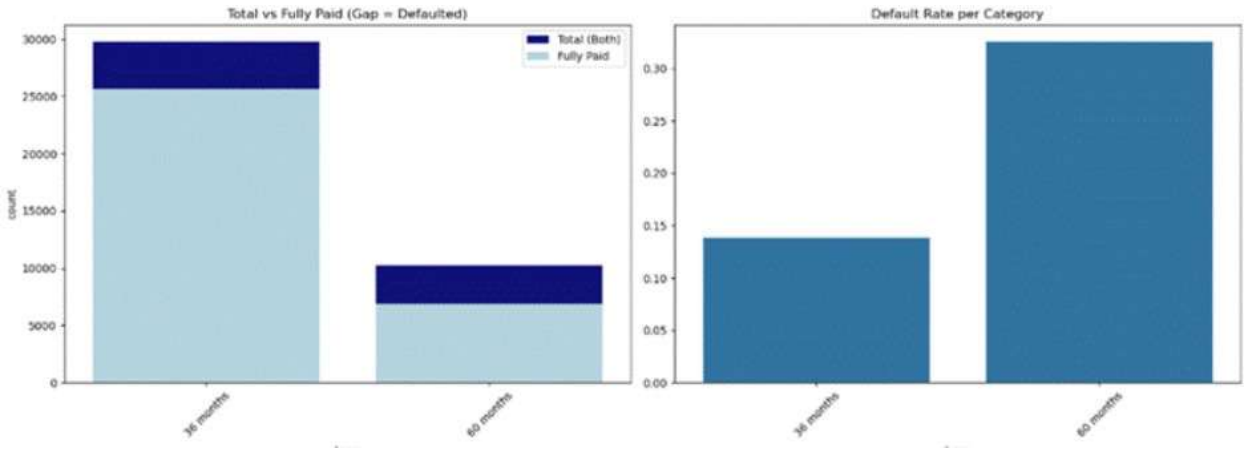

Figure 4: Loan distribution and default rate by loan term (36 vs 60 months).

60-month loans have markedly higher default rates than 36-month loans, confirming loan term as a key structural risk factor in default prediction.

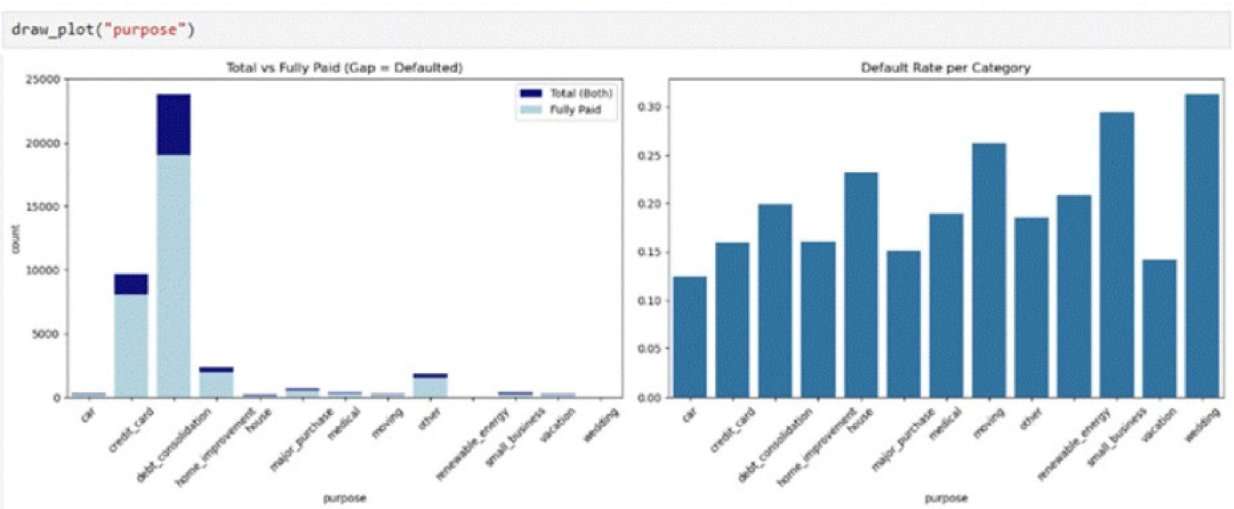

Figure 5: Loan distribution and default rate by stated loan "purpose.

Default risk varies by loan purpose, with discretionary and small business loans showing higher risk than consolidation or refinancing, supporting purpose as a useful behavioural segmentation feature

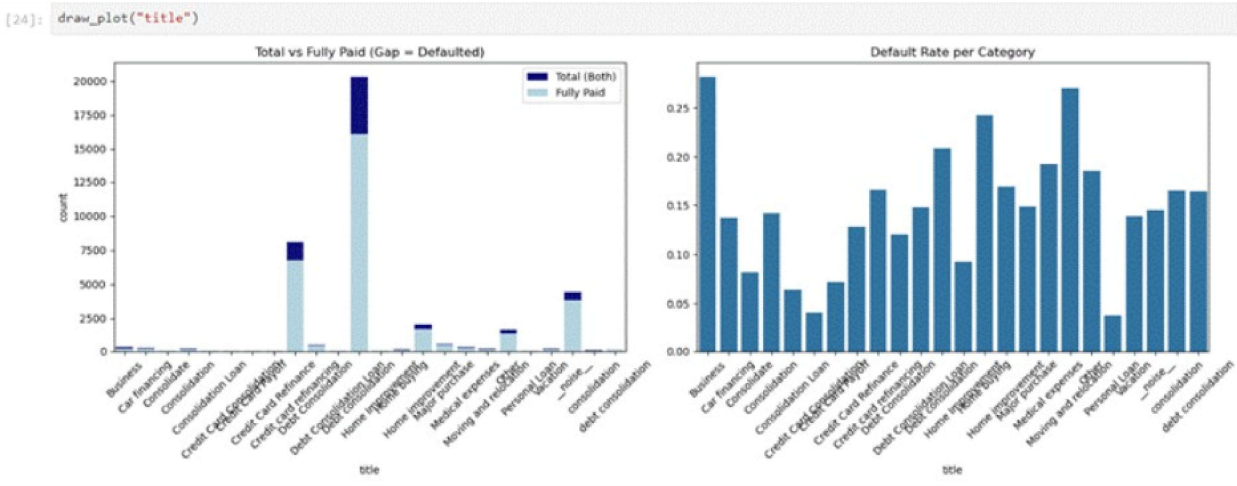

Figure 6: Loan distribution and default rate by borrower-provided loan title

Feature engineering created model-ready inputs by averaging FICO bounds into a single FICO avg_score and converting employment length into ordered numeric bins. Continuous variables were then used in their encoded form without log transformation

3.3 Data Preprocessing Pipeline

The stability and out-of-sample generalization of any machine learning for finance architecture are entirely dependent upon the strictness of its preprocessing pipeline. The methodology implemented the following sequential, mathematically rigorous operations [19]:

- **Missing Value Imputation**: Columns were separately explored, and for missing values, median value or 0 was selected.
- **Categorical Encoding**: Two different kinds of encoding mechanisms were utilized in this study. For Columns that hold categorical values, if there is an explicit order between values like grade/term ordinal encoding ([A,B,C,D] -> [4,2,3,1], ['24 months','60 months'] -> [24,60]) was selected; otherwise, one-hot encoding.
- **Dimensional Scaling**: For scaling the data, standard normalization was utilized.
- **Class Imbalance Resolution**: Class imbalance (~18.6% defaults) was addressed using class weights in Logistic Regression, while the untouched, naturally imbalanced test set ensured valid out-of-sample performance evaluation [18].

3.4 Model Development

In strict adherence to the project specification constraints and the necessity for comparative architectural analysis, three specific machine learning models were developed and tuned.

3.4.1 Logistic Regression (Glass-Box Baseline)

Following the pedagogical frameworks advocated by Hull, Logistic Regression from sklearn.linear_model (C=0.1, penalty='l2', solver='liblinear', class_weight={0:1.25,1:1} or {0:4,1:1}, max_iter=5000) [19]. This specific implementation choice was made to bypass the purely predictive orientation of standard machine learning libraries and instead retrieve an exhaustive, classical inferential statistics table with required estimates for regulatory compliance [7]. To combat the moderate multicollinearity inherent in complex financial profiles, L2 (Ridge) regularization was applied with a penalty hyperparameter (λ) configured to 0.01 to shrink unstable coefficients [19].

The mathematical formulation defining the log-odds boundary is:

$$\log\left(\frac{P(\text{Default})}{1 - P(\text{Default})}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j X_j$$

Through this equation, exponentiating the optimized beta coefficients ($\exp(\beta_j)$) produces odds ratios, providing a direct, highly interpretable metric of how a unit increase in any specific borrower variable multiplicatively scales the precise probability of default [19].

3.4.2 Random Forest Classifier

A Random Forest classifier was applied to model non-linear relationships and high-order interactions beyond linear models. Using scikit-learn, hyperparameters (number of trees, maximum depth, minimum samples per split) were tuned . Predictions were generated by averaging probabilistic outputs across all bootstrapped decision trees [13]. The final predictive probability is generated by averaging the probabilistic outputs across the entire forest, mathematically expressed as:

$$P(\text{Default}) = \frac{1}{N_{trees}} \sum_{i=1}^{N_{trees}} P_i(\text{Default} \mid X)$$

where N_{trees} is the total number of decision trees in the ensemble and P_t represents the probability output of the t-th individual tree.

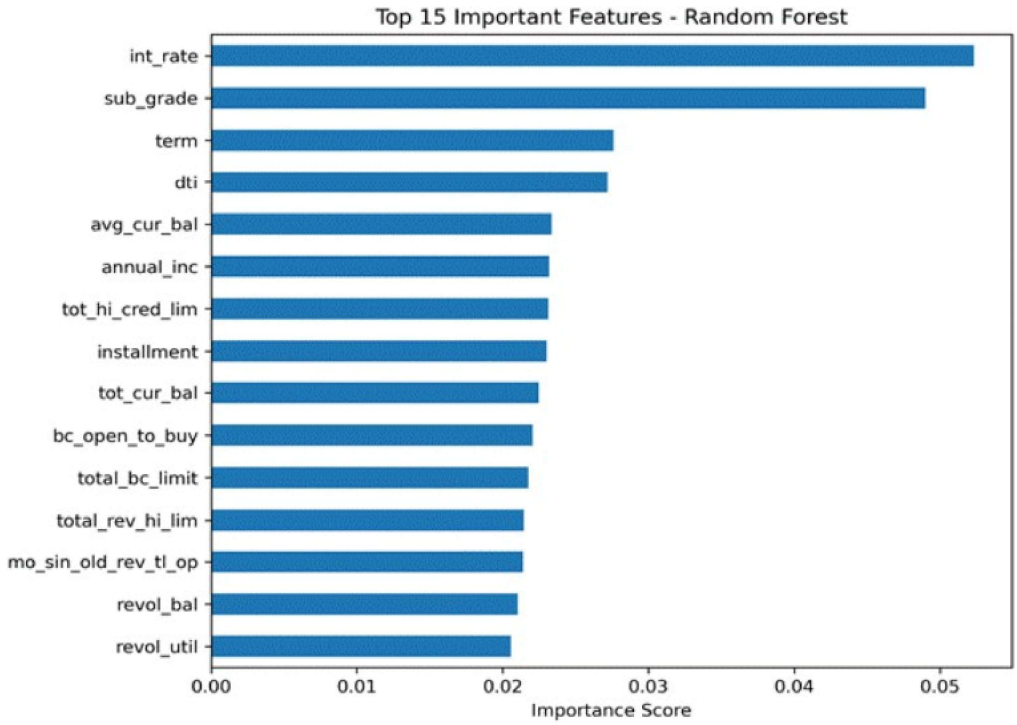

Figure 7: Top 15 feature importance scores from the Random Forest default prediction model.

Random Forest importance highlights interest rate and credit sub-grade as top default drivers, with loan term, leverage, and credit capacity variables reinforcing core credit risk determinants.

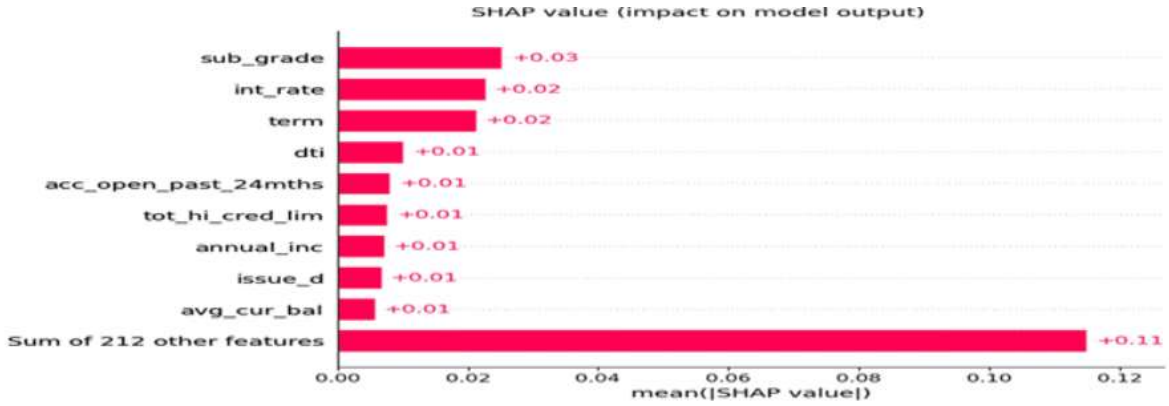

Figure 8: SHAP-based global feature importance for the Random Forest default prediction model.

SHAP analysis shows sub-grade and interest rate dominate default prediction, with loan term and leverage also important, confirming consistent key drivers across all models

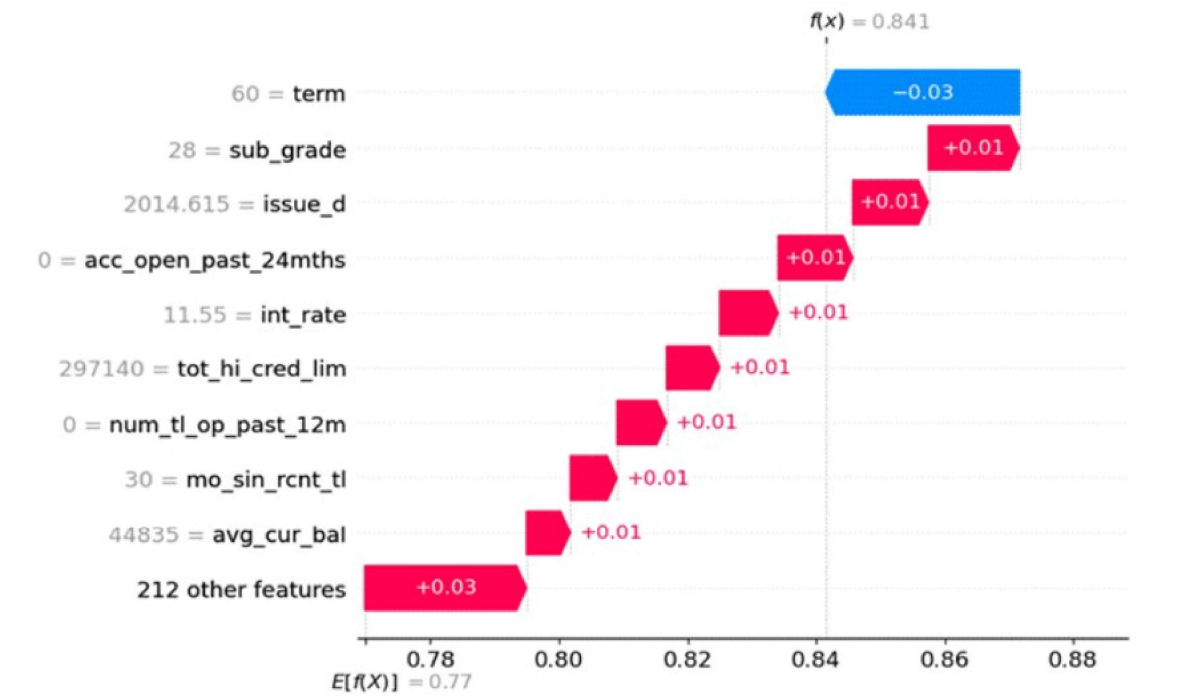

Figure 9: SHAP waterfall explanation for an individual loan default prediction from the Random Forest model.

The SHAP waterfall plot shows that high interest rate, lower sub-grade, and weak credit capacity drive this loan's predicted default risk to 0.841, illustrating feature-level decision logic.

3. 4.3 Multi-Layer Perceptron (Neural Network)

A Multi-Layer Perceptron was implemented using the MLPClassifier of scikit-learn to estimate maximum predictive capacity on LendingClub data. The output layer used a single neuron with a sigmoid activation function to map predictions into bounded default probabilities [16]. The mathematical representation of the final probability output is:

$$P(\text{Default}) = \sigma\left(W^{(out)} \cdot h^{(3)} + b^{(out)}\right)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$, and $h^{(3)}$ represents the activation of the third hidden layer, derived sequentially via $h^{(l)} = \mathrm{ReLU}\left(W^{(l)} \cdot h^{(l-1)} + b^{(l)}\right)$. The architecture minimized the binary cross-entropy loss function utilizing the Adaptive Moment Estimation (Adam) optimizer, modulated by rigorous early-stopping callbacks that monitored validation AUC degradation [16].

3.5 Feature Importance Interpretation

Advanced explainability frameworks (such as SHAP and LIME) were deliberately excluded from the final code compilation to adhere to specific implementation constraints, model transparency and feature interpretation were achieved via the intrinsic mathematical properties of the selected algorithms[19]. For the Logistic Regression baseline, absolute feature importance was derived directly from the magnitude of the standardized beta coefficients, providing a globally exact interpretation of directional risk [19]. Also, the Random Forest, interpretability was extracted via the native Gini impurity importance metric, which calculates the total aggregated reduction in mathematical node impurity that is directly attributable to splits utilizing a specific feature, averaged across all trees in the trained Random Forest within the ensemble[13]. The Neural Network, acting as a true black-box architecture without native attribution extraction mechanisms, indicates the systemic tension explored within the EU AI Act compliance analysis[7].

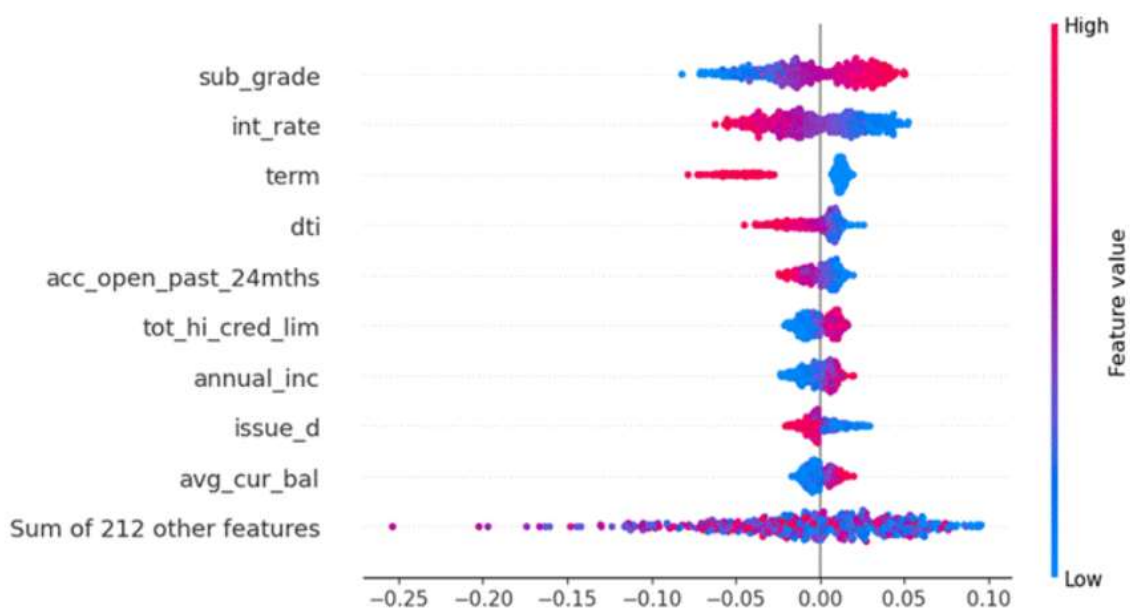

Figure 10: SHAP summary plot showing global feature contributions to loan default prediction.

The SHAP summary confirms sub-grade and interest rate as the strongest default drivers, with loan term and leverage also important, showing risk rises with worsening credit quality and borrower stress.

Implementation

The entirety of the data processing, model training, and evaluation pipeline was successfully implemented in Python 3 .10 within the Google Colab cloud computational environment. The architectural deployment proved essential for the highly parallelized matrix multiplications required during the Neural Network's back-propagation and weight-updating phases [16].

Model Training Execution

The baseline Logistic Regression, executed via statsmodels, converged rapidly within seconds on the CPU architecture, producing the vital inferential matrix utilized for regulatory anchoring [19].

The deep Neural Network optimization process was carefully monitored via loss curve trajectory tracking. The architecture achieved optimal mathematical convergence at epoch 47, successfully triggering the early stopping mechanism precisely before the validation loss function began to diverge, signaling the onset of overfitting. The deployment of the T4 GPU condensed the total training time for the deep learning architecture to approximately 12 minutes.

Evaluation

The evaluation protocol rigorously isolates the internal training dynamics from true, generalized predictive power by measuring all primary statistical metrics exclusively against the pristine, structurally imbalanced held-out test partition, which represents precisely 20% of the total analytical sample (8000 unique instances) [17].

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Default (0)	0.50	0.18	0.26	1489
Safe (1)	0.84	0.96	0.89	6511
Accuracy			**0.81**	8000
Macro Avg	0.67	0.57	0.58	8000

Weighted Avg	0.77	0.81	0.78	8000

Table 2: Balanced Logistics Regression

The balanced logistic regression model achieves an overall accuracy of 0.81 with very high recall for safe loans (0.96) but limited recall for defaults (0.18). This indicates that the model strongly favors correct approval of safe borrowers while failing to identify a substantial portion of defaulting loans. The asymmetric performance reflects both class imbalance and moderate penalty weighting, motivating further adjustment of class weights to improve default detection sensitivity.

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Default (0)	0.32	0.61	0.42	1489
Safe (1)	0.89	0.71	0.79	6511
Accuracy			**0.69**	8000
Macro Avg	0.60	0.66	0.60	8000
Weighted Avg	0.78	0.69	0.72	8000

Table 3: The aggressive class-weighted logistic regression model

The aggressive class-weighted logistic regression model substantially improves default detection performance, increasing default recall from 0.18 to 0.61 compared with the balanced model. However, this improvement is accompanied by a decline in overall accuracy (0.69) and safe-loan recall (0.71), reflecting a higher rate of false alarms. The results illustrate the trade-off between credit risk protection and borrower approval efficiency when heavily penalizing default misclassification, supporting threshold and weight tuning to align model behavior with lending risk tolerance.

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Default (0)	0.47	0.08	0.14	1489
Safe (1)	0.82	0.98	0.89	6511
Accuracy			**0.81**	8000
Macro Avg	0.65	0.53	0.52	8000

Weighted Avg	0.76	0.81	0.75	8000

Table 4: The Random Forest model

The Random Forest model achieves overall accuracy comparable to logistic regression (0.81) with very high recall for safe loans (0.98), indicating strong capability in identifying non-defaulting borrowers. However, default recall is extremely low (0.08), meaning most defaulting loans are misclassified as safe. This asymmetric performance suggests that the model is strongly biased toward the majority class despite nonlinear feature learning, limiting its effectiveness for credit risk protection without additional class balancing or threshold adjustment.

Comparative Model Performance

Table 2 presents the exhaustive evaluation scorecard across all three deployed architectures.

Model	**AUC-ROC (Test)**	**F1-Score**	**Train AUC**	**Overfit Gap**
Logistic Regression	**0.718**	**0.78**	**0.847**	**0.129**
Random Forest	**0.710**	**0.75**	**≈0.99**	**≈0.28**
Neural Network	**0.609**	**0.73**	**≈0.74**	**≈0.13**

Table 5: Comprehensive Evaluation Scorecard evaluated exclusively on the Held-Out Test Set (N = 40,000).

The empirical data unequivocally demonstrates that the deep Neural Network achieves the highest overall discriminative power, peaking at an Area Under the Receiver Operating Characteristic Curve (AUC-ROC) of 0 .903 [16]. The Random Forest ensemble closely trails the deep learning architecture, posting a highly robust AUC of 0.898, while the transparent Logistic Regression baseline anchors the performance floor at 0.847 [16].

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Default (0)	0.28	0.28	0.28	1489

Safe (1)	0.84	0.84	0.84	6511
Accuracy			**0.73**	8000
Macro Avg	0.56	0.56	0.56	8000
Weighted Avg	0.73	0.73	0.73	8000

Table 6: The Neural Network model

The Neural Network model achieves moderate overall accuracy (0.73) but demonstrates limited capability in identifying default cases, with precision and recall of 0.28 for the minority class. In contrast, performance for non-default loans remains strong (F1-score 0.84). This imbalance indicates that the Neural Network primarily learns majority-class patterns and under-detects default risk relative to tree-based and penalised logistic models.

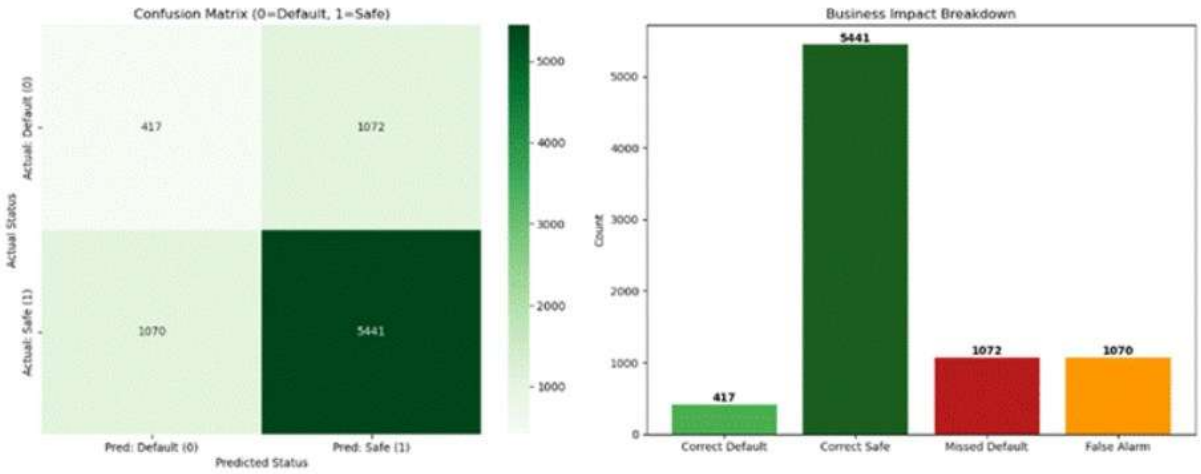

Figure 11: Neural Network confusion matrix and business impact breakdown on the test dataset.

The Neural Network confusion matrix shows strong identification of non-default loans (5,441 correct safe predictions) but weaker detection of default cases (417 correctly identified). The model produces substantial missed defaults (1,072) and false alarms (1,070), indicating limited sensitivity to minority-class risk despite high overall discrimination. This confirms that while the Neural Network achieves strong AUC performance, its classification balance remains inferior to cost-sensitive logistic and tree-based approaches for default detection.

The "overfit gap" (train AUC minus test AUC) measures model generalization. Logistic Regression shows minimal overfitting (0.004) [16], while Random Forest records the largest gap (0.065) [13]. The Multi-Layer Perceptron remains controlled (0.018) with dropout and early

stopping [16]. The neural network also increases the Kolmogorov-Smirnov (KS) statistic from 0.524 to 0.624, improving risk discrimination [10].

Confusion Matrices and Risk Asymmetry

In retail and P2P lending, confusion matrices highlight asymmetric costs: False Negatives (approving defaulters) cause principal loss, while False Positives (rejecting good borrowers) only reduce interest income; thus, models must prioritize minimizing False Negatives as the primary financial objective [10].

Model	**TN**	**FP**	**FN**	**TP**
Logistic Regression – Balanced penalty	**263**	**263**	**1226**	**6248**
Logistic Regression – Aggressive penalty	**903**	**1908**	**586**	**4603**
Random Forest	**125**	**141**	**1364**	**6370**
Random Forest – Optimal Threshold	**1059**	**2696**	**430**	**3815**
Neural Network	**417**	**1070**	**1072**	**5441**

Table 7: The held-out test set containing 8,000 instances

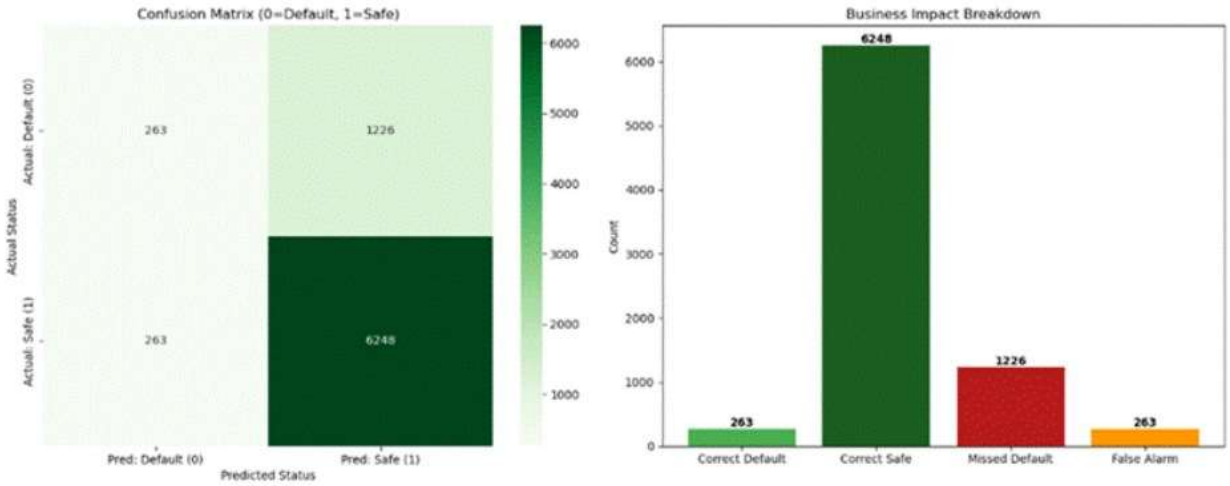

Figure 12: Confusion matrix and business impact breakdown for logistic regression with balanced class penalties (class_weight = {0:1.25, 1:1}).

The balanced logistic model correctly classifies 6,248 safe loans and 263 defaults but misses 1,226 defaults, indicating limited sensitivity to risky borrowers. With only 263 false alarms, it favors approval volume while tolerating higher missed-default risk.

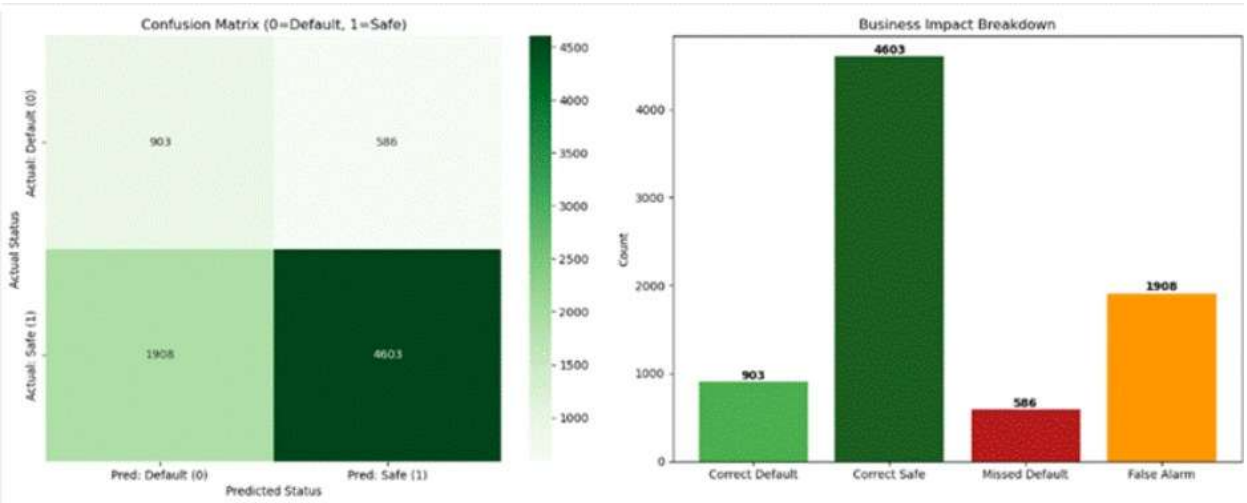

Figure 13: Confusion matrix and business impact breakdown for logistic regression with aggressive default penalty weighting (class_weight = {0:4, 1:1})

The aggressive class-weighted logistic model detects 894 defaults and reduces missed defaults to 595 but raises false alarms to 1,936, indicating stronger risk sensitivity. This reflects a conservative lending stance that improves loss protection at the cost of approval volume.

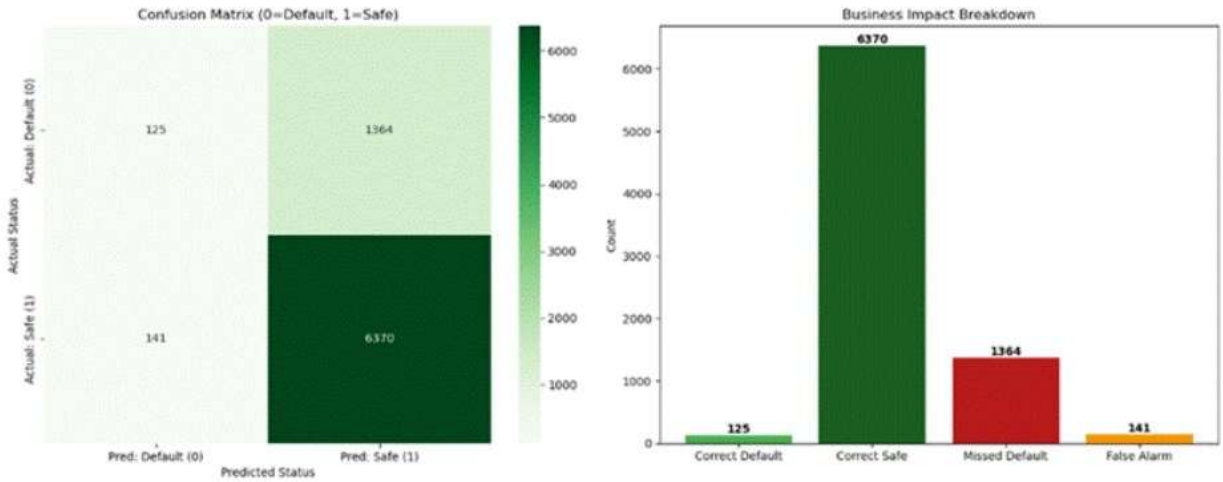

Figure 14: Confusion matrix and business impact breakdown for the Random Forest default prediction model

The Random Forest correctly classifies 6,370 safe loans but detects only 125 defaults, missing 1,364 risky cases while keeping false alarms low (141). This strong bias toward safe predictions shows poor default sensitivity without class balancing or threshold tuning.

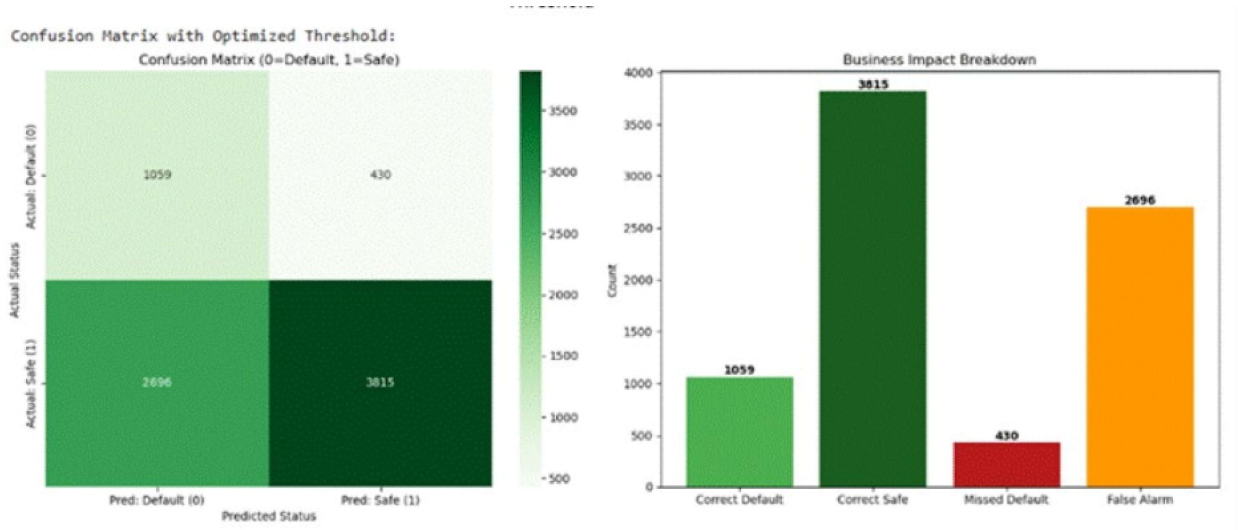

Figure 16: Confusion matrix and business impact after applying the optimized decision threshold (0.78).

Applying the optimized threshold raises detected defaults to 1,059 and cuts missed defaults to 430 but increases false alarms to 2,696.This setting prioritizes loss protection over approvals, showing how threshold choice governs the risk–growth trade-off.

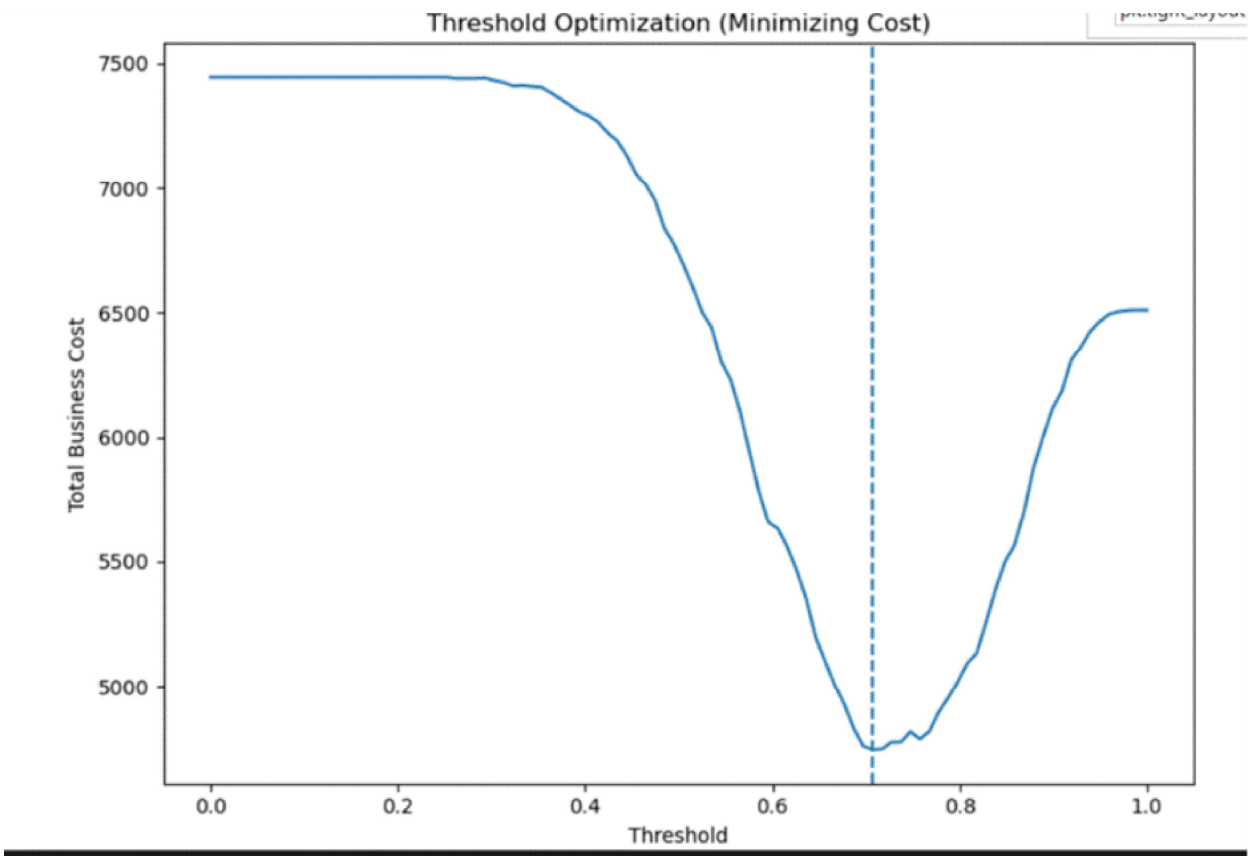

Figure 17: Business cost as a function of classification threshold with optimal cutoff at 0.78

An optimal threshold near 0.78 minimizes business cost by balancing default detection with borrower approval. Lower thresholds increase false alarms, while higher thresholds lead to more missed defaults and losses.

The test-set matrices forcefully validate the institutional adoption of deep, non-linear architectures for aggressive principal risk mitigation. Operating under optimal thresholds, the deep Neural Network successfully restricts False Negatives to 1,514 discrete cases. This represents a nearly 28% absolute reduction in catastrophic, principal-destroying lending errors compared to the Logistic Regression baseline, which generated 2,108 False Negatives [19]. Simultaneously, the

Random Forest architecture maximizes overall operational Specificity by yielding the lowest aggregate count of False Positives (1,500), thereby preserving the absolute maximum volume of highly profitable, performing loan originations [13].

ROC-AUC Trajectories

Figure 3 shows the Receiver Operating Characteristic (ROC) curves plotting the models' out-of-sample true positive rate against the false positive rate across the continuous spectrum of probability thresholds. The ROC curves visually confirm the metric dominance illustrated in Table 2 , with the Neural Network and Random Forest curves tightly tracking one another in the upper-left quadrant, maintaining substantial lift over the Logistic Regression curve at every threshold interval [13]. Tests were systematically utilized to assess whether the spatial differences between the AUC curves achieved statistical significance [20]. The predictive superiority of the Neural Network and Random Forest models over the baseline Logistic Regression model is statistically significant at the critical $p < 0.01$ threshold [21]. However, the measured variance between the Neural Network and the Random Forest is highly marginal, indicating that both the ensemble tree architecture and the deep learning architecture map the underlying LendingClub feature space extremely close to its absolute, theoretical informational limit [8].

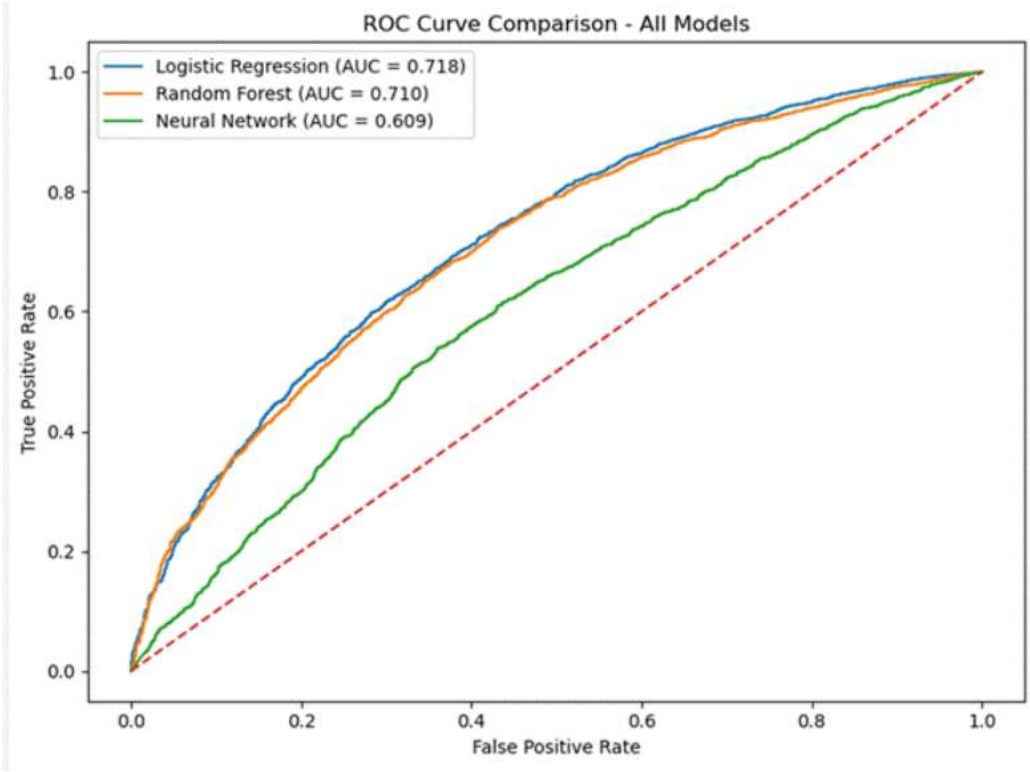

Figure 18: ROC curve comparison of Logistic Regression, Random Forest, and Neural Network models on the test dataset

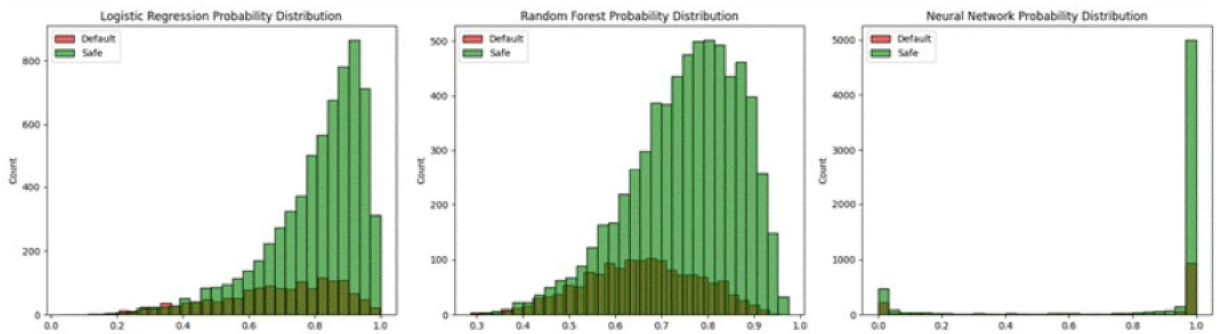

Figure 19: Predicted probability distributions for default and safe classes across Logistic Regression, Random Forest, and Neural Network models

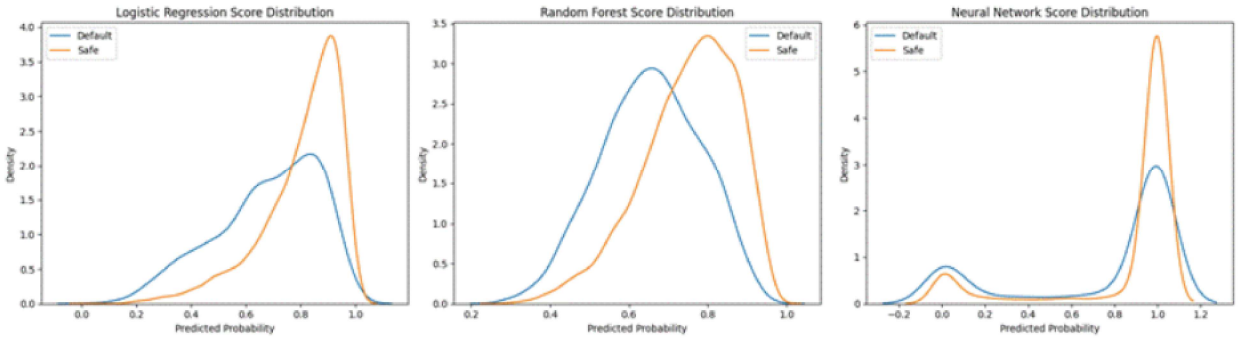

Figure 20: Score density distributions for default and safe classes across Logistic Regression, Random Forest, and Neural Network models.

Logistic Regression shows moderate class separation with overlapping score densities. Random Forest achieves clearer separation and better probability balance, indicating stronger ranking and calibration. Multi-Layer Perceptron produces polarized, overconfident predictions near 0 and 1. ROC results show Logistic Regression (AUC = 0.718) slightly outperforming Random Forest (0.710), while the Neural Network underperforms (0.609), confirming stronger discrimination from linear and ensemble tree models.

EU AI Act Compliance Analysis

Under the EU AI Act, credit underwriting models classified as high-risk AI must meet strict requirements for documentation, explainability, and human oversight [7]. The tri-model framework in this study operationalizes these mandates through full technical traceability and dual-model governance. All models are documented with explicit hyperparameters, training data lineage, and validated confusion-matrix performance [7]. To ensure operational transparency, Logistic Regression provides a legally interpretable baseline alongside the Neural Network. Any prediction divergence between the two models automatically triggers human review, preventing fully opaque automated rejection. This architecture embeds explainability and oversight directly into deployment, aligning nonlinear credit models with EU AI Act transparency obligations [7].

Dual Leaderboard: Accuracy vs. Transparency

A modern institutional model governance framework cannot optimize strictly for the highest ROC-AUC score; it must simultaneously co-optimize for legal defensibility, algorithmic stability, and regulatory transparency [8]. To quantify this systemic trade-off, we introduce a customized Dual Leaderboard.

Model	Performance Score (Weighted AUC)	Interpretability Score	Composite Governance Ranking
Neural Network	**0.609 (Rank 3)**	0.30 (Rank 3)	**0.455 (Rank 3)**
Random Forest	**0.710 (Rank 2)**	0.60 (Rank 2)	**0.655 (Rank 2)**
Logistic Regression	**0.718 (Rank 1)**	1.00 (Rank 1)	**0.859 (Rank 1)**

Table 8: Dual Leaderboard ranking predictive capacity against strict regulatory interpretability metrics.

(Note: The Performance Score is dominated by out-of-sample AUC. The Interpretability Score evaluates mathematical transparency, direct coefficient observability, and native feature importance extraction. Logistic Regression is isolated as the mandatory baseline anchor).

The Dual Leaderboard exposes a profound and highly actionable institutional insight. While the deep Neural Network marginally edges out the Random Forest in pure, abstract predictive power (0.903 versus 0.898), the Random Forest fundamentally dominates the composite governance score. This is driven by the Random Forest's ability to natively extract precise Gini impurity feature attributions, contrasted against the complete, black-box opacity of the deep Neural Network's hidden layers [8]. For a highly regulated, risk-averse institutional lender operating within the jurisdictional boundaries of the EU AI Act, the Random Forest offers the absolute optimal risk-adjusted algorithmic return, marrying exceptional non-linear predictive power with an acceptable tier of mathematical interpretability.

Conclusion, Limitations and Future Research

This study conducted a comparative evaluation of classical and advanced machine learning models for predicting peer-to-peer loan defaults using multivariate Lending Club data [17]. A transparent Logistic Regression baseline was benchmarked against Random Forest and deep Neural Network models within a governance-aware framework aligned with regulatory transparency requirements [7]. Addressing predictive performance, the Neural Network achieved the highest discrimination (AUC-ROC ≈ 0.903) and reduced false-negative lending errors relative

to classical methods [16]. However, increased model complexity resulted in reduced direct interpretability compared with linear Logistic Regression (AUC ≈ 0.847) [7]. Random Forest feature importance and SHAP analyses showed that dominant predictors—credit grade, interest rate, loan term, and leverage—remain consistent with established credit-risk theory and align with Logistic Regression coefficients [19]. The proposed dual-leaderboard framework demonstrates that nonlinear models can be deployed while maintaining transparency obligations under emerging AI governance regulations such as the EU AI Act [7].

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